

Code	Explanation
<code>import os import cv2</code>	Imports the required libraries.
<code>cv2.NamedWindow('image')</code>	Opens a window to show the image.
<code>while True:</code>	
<code>os.system('fswebcam -r 1024x768 -S 3 --jpeg 50 --save / home/pi/Desktop/image.jpg')</code>	Reads the image using the fswebcam utility and saves it to the desktop as image.jpg.
<code>frame = cv2.imread("/home/pi/Desktop/ image.jpg")</code>	Loads the image captured from the webcam.
<code>gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)</code>	Converts the image to grayscale.
<code>faces = faceCascade.detectMultiScale(gray, scaleFactor=1.1, minNeighbors=5, minSize=(10, 10), flags = cv2.cv.CV_HAAR_SCALE_IMAGE)</code>	Detects all faces in the image.
<code>for (x, y, w, h) in faces: cv2.rectangle (frame, (x, y), (x+w, y+h), (0, 79, 255), 2)</code>	For every face detected, it draws a light blue border to show every face detected.
<code>cv2.ShowImage('image',frame)</code>	Displays the image.
<code>k=cv2.WaitKey(0);</code>	Waits to see if the user presses a key.
<code>if(k == 'ESC'): break</code>	Exits the loop if the user presses the [Escape] key.
<code>cv2.DestroyWindow('image')</code>	Closes the window.

Next, in the terminal, type:

- `Python face _ detect.py`

This should open a window that detects faces.

Get an LED to blink

In this section, to get started, we'll use a simple circuit that can switch an LED ON or OFF. You can take the help of an electronics expert to extend this project to turn appliances on and off in your home.